

# **Complexity Bounds for Dirichlet Process Slice Samplers**

Beatrice Franzolini

*Joint work with:*  
**Francesco Gaffi**



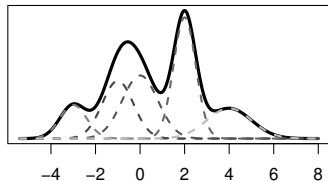
# DP-based models

$$\begin{array}{ccc}
 Y \mid \eta, \mathbf{z}_n \sim \underbrace{\mathcal{L}(\mathbf{z}_n, \eta)}_{\substack{\text{likelihood} \\ \text{depends on} \\ \mathbf{z}_n = (z_1, \dots, z_n)}} & z_i \mid G \stackrel{iid}{\sim} \underbrace{G(dx) \stackrel{a.s.}{=} \sum_{k=1}^{\infty} \pi_k \delta_{\phi_k}(dx)}_{\substack{\text{discrete random} \\ \text{probability measure}}} & \underbrace{G \sim \text{DP}(\alpha, P_0)}_{\text{prior}}
 \end{array}$$

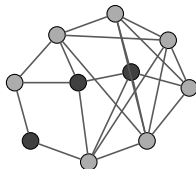
$$G \sim \text{DP}(\alpha, P_0) \implies \phi_k \stackrel{iid}{\sim} P_0, \quad V_k \stackrel{iid}{\sim} \text{Beta}(1, \alpha) \quad \pi_k = V_k \prod_{\ell < k} (1 - V_\ell)$$

$$\implies \rho_n \text{ partition of } [n] = \{1, \dots, n\} \text{ defined by } i \sim j \text{ if and only if } z_i = z_j$$

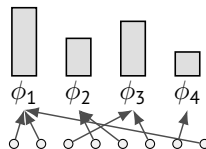
Mixture models



Stochastic block models



Species sampling models



# MCMC sampling in DP-based models

$$Y \mid \eta, \mathbf{z}_n \sim \underbrace{\mathcal{L}(\mathbf{z}_n, \eta)}_{\substack{\text{likelihood} \\ \text{depends on} \\ \mathbf{z}_n = (z_1, \dots, z_n)}} \quad z_i \mid G \stackrel{iid}{\sim} \underbrace{G(dx) \stackrel{a.s.}{=} \sum_{k=1}^{\infty} \pi_k \delta_{\phi_k}(dx)}_{\substack{\text{discrete random} \\ \text{probability measure}}} \quad \underbrace{G \sim \text{DP}(\alpha, P_0)}_{\text{prior}}$$

## Marginal algorithms

(Neal, 2000)

$$\text{pr}(z_i = z | \text{rest}) \propto \begin{cases} n_h(\mathbf{z}_n^{-(i)}) \times \mathcal{L}(\mathbf{z}_n, \eta) & \text{if } z \in \mathbf{z}_n^{-(i)} \\ \alpha \times \int \mathcal{L}(\mathbf{z}_n, \eta) P_0(dz) & \text{if } z = \text{new} \end{cases}$$

- ✓ Target the posterior
- ✗ Force one-at-a-time updates
- ✗ Bookkeeping

## Blocked Gibbs samplers

(Ishwaran and James, 2001)

$$\text{pr}(z_i = z | \text{rest}) \propto \pi_{k(z)} \times \mathcal{L}(\mathbf{z}_n, \eta)$$

- ✗ Target an approximation of the posterior
- ✓ Allow joint updates
- ✓ Easy/No bookkeeping

# Non-vanishing truncation bias of blocked Gibbs samplers

## Exponential accuracy of truncated blocked Gibbs samplers (Ishwaran and James, 2002)

Let  $L$  be the truncation level, i.e.,  $G$  is approximated by  $\sum_{\ell=1}^L \pi_{\ell} \delta_{\phi_{\ell}}$  and  $\rho_n$  the partition induced by ties in  $z_i$  in a Normal mixture model

$$\int_{\mathbb{R}^n} \sum_{\rho_n} \left| \underbrace{p_L(\rho_n \mid \mathbf{y}_{1:n})}_{\text{target of the algorithm}} - \underbrace{p_{\infty}(\rho_n \mid \mathbf{y}_{1:n})}_{\text{posterior of the partition}} \right| \underbrace{m_{\infty}(\mathbf{y}_{1:n})}_{\text{prior predictive}} d\mathbf{y}_{1:n} = O(4 n e^{-(L-1)/\alpha}).$$

Extensions in Ishwaran and Zarepour (2002); Campbell et al. (2019); Li and Campbell (2021).

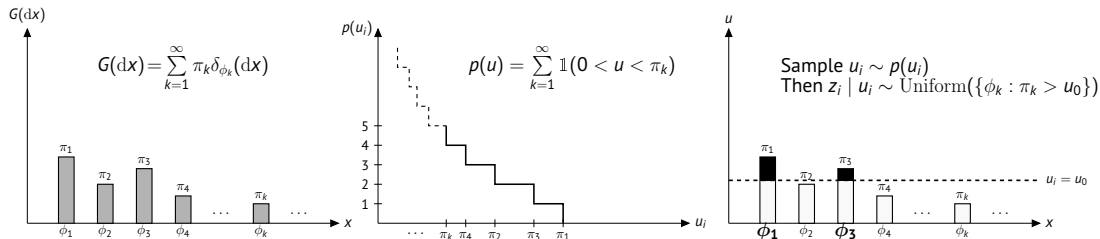
Two main limitations of the result:

✗ It holds **under the generative model**

✗ Trivially: unless  $L = n$ ,  $p_L(\rho_n \mid \mathbf{y}_{1:n}) = 0$  on all partitions with more than  $L$  clusters.

# Slice sampling (Neal, 2003; Walker, 2007)

$$z_i \sim G(dz) \quad \rightarrow \quad (u_i, z_i) \sim \mathbb{1}(0 < u < G(dz))$$



- ✓ Target the posterior
- ✓ Allow joint updates
- ✓ Easy/No bookkeeping

# Slice sampling (Neal, 2003; Walker, 2007; Ge et al., 2015)

## Posterior slice sampling in DP-based models

1. Sample  $G$ , i.e.,  $(\pi_1, \pi_2, \dots)$  and  $(\phi_1, \phi_2, \dots)$
2. Sample  $u_i$  from  $p(u_i | z_i = \phi_k) = \text{Uniform}(0, \pi_k)$   $i = 1, \dots, n$
3. Sample  $z_i$  from  $p(z_i | u_i) = \text{Discrete}(\phi_k : \pi_k > u_i, \text{ with } w_k \propto \text{Lik}(Y; z_i = \phi_k))$   $i = 1, \dots, n$

How many  $(\pi_h, \phi_k)'s$  do we need to sample at step 1?

→ Step 2. requires weights for all currently allocated components.

$$\underbrace{(\pi_1, \dots, \pi_H, \pi^*)}_{H=\text{number of clusters}} \sim \text{Dirichlet}(n_1, \dots, n_H, \alpha) \quad (\text{Ge et al., 2015})$$

→ Step 3. requires determination of the slice set  $A_i = \{\phi_k : \pi_k > u_i\}$ , let  $u_{\min} = \min_i u_i$

$$\forall i, A_i \subseteq \left\{ \phi_1, \dots, \phi_K : \sum_{h=K+1}^{\infty} \pi_h < u_{\min} \right\} \quad (\text{Walker, 2007})$$

## Dynamic truncation level $K$

$$K = \min \left\{ k \geq H : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\} \quad \begin{array}{l} \checkmark \text{ At each iteration, } K \text{ components are enough and } K \text{ is finite.} \\ \times \text{ Time complexity is driven by at most } K \times n \text{ likelihood evaluations.} \end{array}$$

## About K

$$K = \min \left\{ k \geq H : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\}$$

✗ If  $u_{\min} \approx 0$ ,  $K$  can be arbitrarily large.

The diagram shows the expression  $\mathbb{P}_{\pi}(\exists s \leq T : C_s > C)$  with three arrows pointing to its components: *stationary distribution* points to  $\mathbb{P}_{\pi}$ , *any finite threshold* points to  $C$ , and *computational cost of iteration s (in some unit of measure)* points to  $C_s$ . To the right of the expression is a horizontal arrow pointing to the number 1, with  $T \rightarrow \infty$  written below it.

$$\mathbb{P}_{\pi}(\exists s \leq T : C_s > C) \xrightarrow{T \rightarrow \infty} 1$$

→ **No deterministic bound on per-iteration time exists.**

**Informally:** To assess actual scalability, we need to quantify the “probability of exceeding a threshold” and how this varies with  $n$ .

## About $K$ (from now $K_n$ )

A related quantity:  $K(x) = \min \left\{ k \geq 1 : \sum_{h=k+1}^{\infty} \underbrace{\pi_h}_{\text{from the prior}} < x \right\} \quad x \in (0, 1)$

From Muliere and Tardella (1998)

$$K \mid x \sim 1 + \text{Poisson}(-\alpha \log x)$$

But at **each MCMC iteration** (in the improved slice sampler of Ge et al., 2015)

$$K_n = \min \left\{ k \geq H_n : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\}$$

where:

$H_n$  is the number of clusters of the partition  $\rho_n$  visited by the chain

$\pi_h$ 's depend on the partition  $\rho_n$  visited by the chain

$u_{\min}$  depends on the  $(\pi_h)_{h \geq 1}$  and on the partition  $\rho_n$  visited by the chain

$\rho_n$  depends on the specific data

# Survival probability of $u_{\min}$

**Proposition: Merging two clusters increases the survival probability of  $u_{\min}$**

If  $\rho_n^{(r \oplus s)}$  is the partition obtained from  $\rho_n$  by merging two (distinct) clusters  $r$  and  $s$  into a single cluster and leaving the other clusters unchanged, then, for any  $x \in (0, 1)$

$$\Pr(u_{\min} > x \mid \rho_n^{(r \oplus s)}) \geq \Pr(u_{\min} > x \mid \rho_n).$$

**Corollary: Singleton partition yields the lowest survival probability of  $u_{\min}$**

Let  $\rho_n^{\text{sing}}$  denote the singleton partition of  $[n]$ , i.e., the partition with  $n$  clusters of size 1. Then, for every  $x \in (0, 1)$  and every partition  $\rho_n$ ,

$$\Pr(u_{\min} > x \mid \rho_n) \geq \Pr(u_{\min} > x \mid \rho_n^{\text{sing}})$$

# Main result

## Theorem: High-probability bound on dynamic truncation level (Franzolini and Gaffi, 2026)

Let  $K_n$  and  $H_n$  be respectively the truncation level and the number of occupied clusters at a given iteration when run for  $n$  input data.

Then, for every  $\delta \in (0, 1)$  and any  $n \geq 2$

$$\Pr(K_n - H_n \leq C_\delta \log n \mid \rho_n) \geq 1 - \delta, \quad \forall \rho_n$$

with  $C_\delta = B_\alpha^{(1)} + B_\alpha^{(2)} \log(1/\delta)$ , and  $B_\alpha^{(1)}, B_\alpha^{(2)} \asymp \alpha$ .

In particular, as  $n \rightarrow \infty$ ,  $K_n - H_n = O_{\mathbb{P}}(\log n)$  and, in the worst-case scenario of  $H_n = O(n)$ ,  $K_n = O_{\mathbb{P}}(n)$ .

*“In the worst case, the slice sampler requires  $O(n)$  active components with high probability.”*

## Proof sketch

$$K_n = \min \left\{ k \geq H_n : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\} = \min \left\{ k \geq H_n : \pi^* \times \prod_{i=H_n+1}^k (1 - V_i) < u_{\min} \right\}.$$

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■ Define  $K_n^0(x) = \min \left\{ k > H_n : \prod_{i=H_n+1}^k (1 - V_i) < x \right\}$

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■  $\Pr(K_n - H_n > C_\delta \log n \mid \rho_n) \leq \boxed{\Pr(K_n^0(x) > H_n + C_\delta \log n \mid \rho_n)} + \boxed{\Pr(K_n > K_n^0(x) \mid \rho_n)}$

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■ Show that  $\boxed{\Pr(K_n^0(x) \geq 1 + H_n + 3\alpha \log(1/x) + \log(2/\delta) \mid \rho_n) \leq \frac{\delta}{2}} \quad (1)$

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■ Show that  $\boxed{\Pr(K_n > K_n^0(x) \mid \rho_n) \leq \Pr(u_{\min} \leq x \mid \rho_n) \leq n(\alpha + n - 1)x[1 + \log(1/x)]} \quad (2)$

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■ Choose  $x = x(n, \delta) \in (0, 1)$  such that, for any  $n \geq 2$ , the lower-bound in the probability in (1) is upper-bounded by  $H_n + C_\delta \log n$ , and the r.h.s. of (2) is upper-bounded by  $\delta/2$ .

## A few additional results

### Corollary: exponential tails of the slice overhead

Let  $(K_n, H_n)$  be defined as in the main Theorem.

There exist constants  $B_\alpha^{(1)}, B_\alpha^{(2)} > 0$  such that for all  $n \geq 2$  and all  $t \geq 0$ ,

$$\Pr \left( \frac{K_n - H_n}{\log n} > B_\alpha^{(1)} + B_\alpha^{(2)} t \mid \rho_n \right) \leq e^{-t}, \quad \forall \rho_n.$$

### Corollary: Almost-sure control of super-logarithmic slice overheads with infinite data coupling

Let  $(K_n, H_n)$  be defined as in the main Theorem.

There exists a constant  $D_\alpha > 0$  such that

$$\Pr \left( \limsup_{n \rightarrow \infty} \left\{ K_n - H_n > D_\alpha \log \frac{1}{\delta_n} \log n \right\} \right) = 0$$

for any summable sequence  $(\delta_n)_{n \geq 1} \subset (0, 1/2)$ .

## A few additional results

### Proposition: tightness of the logarithmic slice-overhead bound

Let  $(K_n, H_n)$  be defined as in the main Theorem.

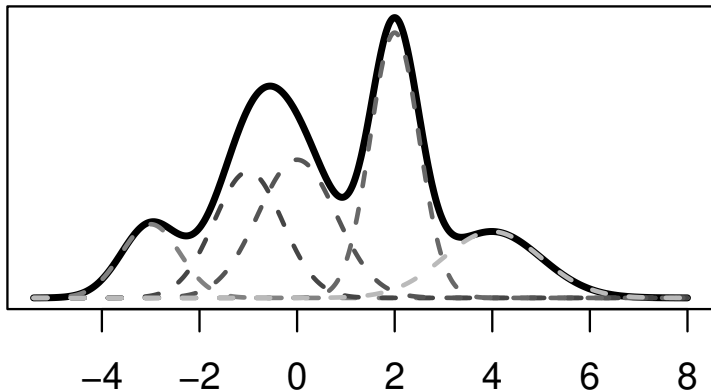
There exist constants  $c_\alpha, \eta_\alpha > 0$  such that for all sufficiently large  $n$ ,

$$\Pr \left( K_n - H_n \geq c_\alpha \log n \mid \rho_n^{\text{sing}} \right) \geq \eta_\alpha.$$

Consequently, for any deterministic sequence  $r_n = o(\log n)$  and any  $M > 0$ ,

$$\liminf_{n \rightarrow \infty} \sup_{\rho_n} \Pr (K_n - H_n > Mr_n \mid \rho_n) > 0.$$

## Practical consequences for mixture models



# MCMC alternatives comparison

$$\text{for } i \in \{1, \dots, n\} \quad Y_i \mid z_i \stackrel{\text{ind}}{\sim} \mathcal{N}(z_i, \sigma^2), \quad z_i \mid G \stackrel{\text{iid}}{\sim} G, \quad G \sim \text{DP}(\alpha, P_0),$$

|                                         | CRP           | BGS-L       | BGS-n         | Slice                      |
|-----------------------------------------|---------------|-------------|---------------|----------------------------|
| Scalability by posterior cluster growth |               |             |               |                            |
| $H_n = O(n)$                            | $O(n^2)$      | $\Theta(n)$ | $\Theta(n^2)$ | $O_{\mathbb{P}}(n^2)$      |
| $H_n = O(\log n)$                       | $O(n \log n)$ | $\Theta(n)$ | $\Theta(n^2)$ | $O_{\mathbb{P}}(n \log n)$ |
| $H_n = O(1)$                            | $O(n)$        | $\Theta(n)$ | $\Theta(n^2)$ | $O_{\mathbb{P}}(n \log n)$ |
| Exact posterior partition target        | ✓             | ✗           | ✗             | ✓                          |
| No hard-threshold bias                  | ✓             | ✗           | ✓             | ✓                          |
| No bookkeeping                          | ✗             | ✓           | ✓             | ✓                          |
| Joint Updates                           | ✗             | ✓           | ✓             | ✓                          |

# Numerical study

Competing algorithms:

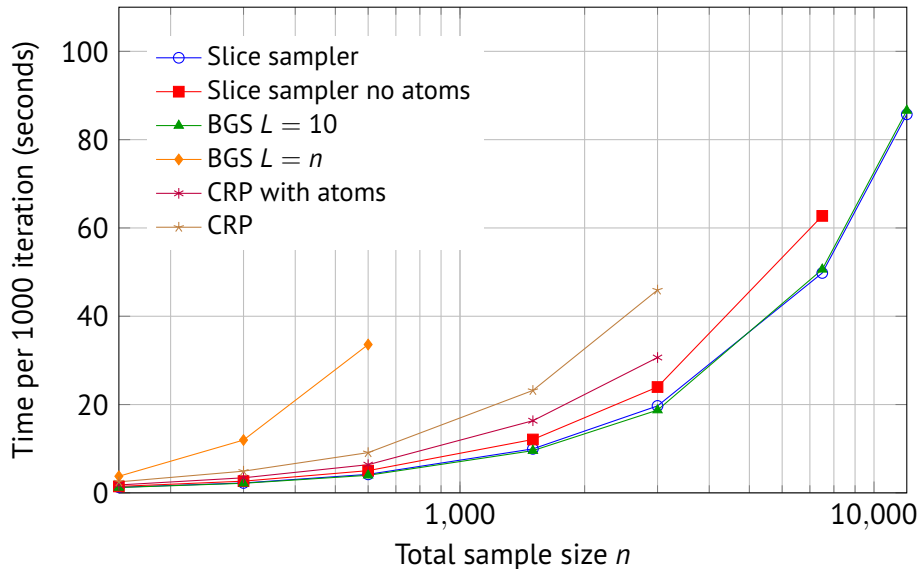
- ★ Slice sampler
- ★ Slice sampler with no atoms
- ★ BGS  $L = 10$
- ★ BGS  $L = n$
- ★ Marginal CRP
- ★ Marginal CRP with atoms

Simulated data scenario: Three clusters of equal size

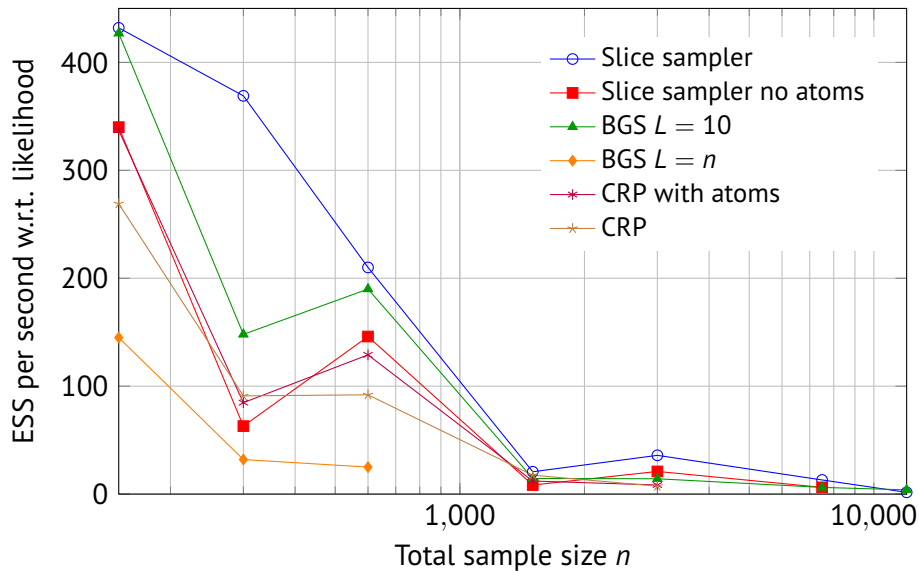
$$y_i \sim \mathcal{N}(\phi_{z_i}, \tau^{-1})$$

with true cluster means  $\phi = (-3, 0, 3)$  and precision  $\tau = 1$

## Wall-clock time



# Mixing



# Numerical study

Competing algorithms:

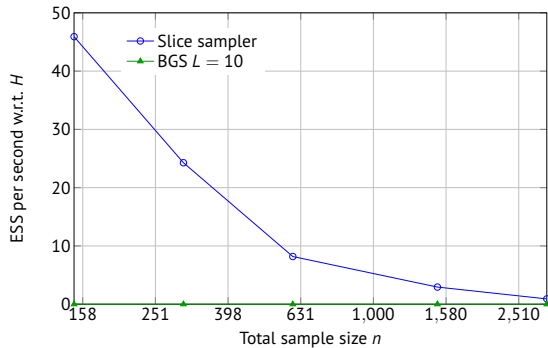
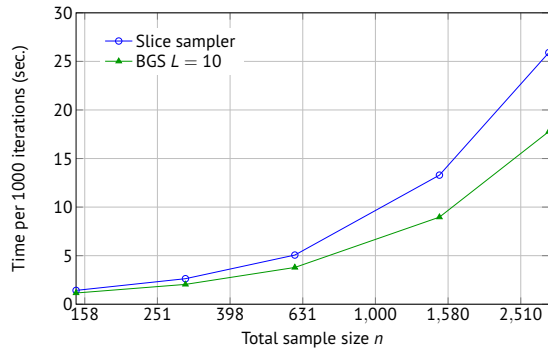
- ★ Slice sampler
- ★ BGS  $L = 10$

Simulated data scenario: Perturbed Zipf

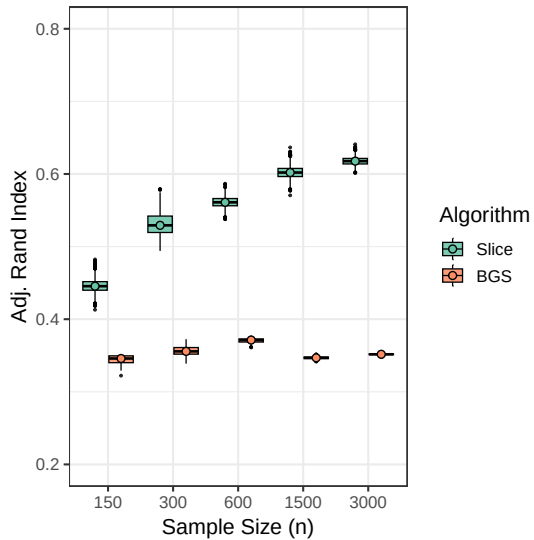
$$y_i \sim \mathcal{N}(\phi_{z_i}, \tau^{-1})$$

$$\phi_{z_i} = 3 \times z_i \quad p(z_i) \propto z_i^{-0.5} \quad \text{for } z_i \in \{1, \dots, 50\}$$

# Wall-clock time and mixing



# Inferential performance



# Conclusions and future directions

## Take-home message

- Slice samplers for DP-based models have **unbounded** per-iteration cost:

$$K_n = \min \left\{ k \geq H_n : \sum_{h=k+1}^{\infty} \pi_h < u_{\min} \right\}.$$

- Nevertheless, for **any dataset**, the slice overhead is **small with high probability**:

$$K_n - H_n = O_{\mathbb{P}}(\log n), \quad \text{uniformly over all visited partitions } \rho_n.$$

## Future directions

- Extend the bounds to **adaptive DP models**, where the concentration parameter  $\alpha$  is learned from the data.
- Study **richer BNP priors**: Pitman–Yor processes (Canale et al., 2022) and general species sampling processes (Mena et al., 2025).
- Develop guarantees for **structured models**: hierarchical DPs, sticky constructions, and dependent or temporally evolving partitions.

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